



VanSafe

Safer school-van rides.

A trust, tracking & safety platform for Pakistan's school vans — and the city-scale mobility data it unlocks.

Civic Innovation Hackathon · Lahore & Karachi

Mohammad Ibrahim Nasir · June 2026

THE PROBLEM

It started at my own front door.

Every school morning my parents faced the same anxiety — and so does every parent who sends a child off in a stranger's van.

- **Drivers go unresponsive** — phones ring out, even mid-route, leaving parents with no idea where their child is.
- **Heavy traffic** turns a 20-minute run into an hour; no live location to plan around it.
- **Extreme weather** — 45°C+ heat and monsoon flooding make delays dangerous, not just inconvenient.
- **No way to verify trust** — choosing a driver is word-of-mouth and hope, with no records, no reviews, no accountability.

WHY IT MATTERS

A safety problem at national scale.

~27,500

road-traffic deaths a year
in Pakistan

Millions

of children commute
daily in informal vans &
wagons

45°C+

summer cabin heat;
monsoon flooding shuts
roads

Peak

city congestion lands on
school drop-off & pickup
hours

Informal school transport is everywhere, essential — and almost entirely **unmonitored and unverified**.

Indicative figures. Sources: WHO Global Status Report on Road Safety; national press & meteorological reports.

THE SOLUTION

VanSafe turns a blind, informal ride into a tracked, trusted one.



Find a trusted van

Browse, filter and AI-match verified drivers by school, area, seats & rating.



Track it live

Real-time GPS on a map, departure & arrival alerts, anomaly detection.



Stay informed

Ask a WhatsApp bot "where is the van?" in English or Urdu — anytime.

Built as a full working product — not a mockup.

Built for parents and drivers.

For parents

- Find & compare trusted vans
- Track the ride live, share location instantly
- Mark a child absent; get safety alerts
- Chat the bot in English or Urdu

For drivers

- A verified profile parents can trust
- One-tap location sharing each morning
- Route optimization & fuel-savings
- Reviews that build a real reputation

WHAT WE BUILT

A complete, working platform.

AI van matching

Gemini ranks the safest, closest, best-rated vans.

Live GPS tracking

30-sec pings on a map with route history.

WhatsApp bot

Twilio + in-app simulator, EN/Urdu intent.

Proactive alerts

Departed / arrived / stationary anomaly detection.

Verification & reviews

CNIC & vehicle docs, AI-summarised reviews.

Vehicle-safety catalog

Real PK vans; official-capacity over-load warnings.

Route optimization

Nearest-neighbour + Google Directions, daily attendance.

Fuel-savings dashboard

Rupees saved this month, 30-day trend.

Pin-drop locations

Map picker for home & pickup, like delivery apps.

Next.js + React

Supabase (Postgres · Auth · Realtime)

Google Gemini

Twilio WhatsApp

Google Directions

Leaflet · OpenStreetMap

Finding the right van.

- 1 **Browse & filter** drivers by area, school, seats and rating.
- 2 **AI matchmaking** — enter the school & area, get a ranked shortlist with reasons.
- 3 **Add a child** and **pin the pickup point** on the map.

No phone-tree, no guesswork — a trusted match in under a minute.

Driver on the phone, parent on the screen.

- 1 Driver taps **Start Route** on their phone — GPS sharing begins.
- 2 Parent's map **updates in real time** as the van moves.
- 3 A "**van departed**" **WhatsApp alert** reaches the parent automatically.

The exact anxiety from slide 2 — answered, live.

"Where is the van?" — وین کہاں ہے؟

In-app simulator

Same bot logic, no setup — answers with a live location & map link.

Real WhatsApp (Twilio)

Text the number — the running demo van's live position comes back to your phone.

Meets parents where they already are — bilingual, instant, zero app-switching.

Why drivers opt in.

- **Credibility, not word-of-mouth** — a verified profile & real reviews win parents a stranger never could.
- **More bookings** — visibility in search & AI-match fills empty seats.
- **Money saved** — route optimization cuts distance & fuel; a dashboard shows rupees saved each month.

Parents bring trust, drivers bring supply — and every trip quietly produces data.

Every ride is an anonymised data point.

Ease school-hour congestion

The most-used school-run **routes & corridors**, mapped. Shared with traffic authorities to re-time signals and relieve the streets that choke at 7am & 1pm.

Faster emergency response

Route density & timing show where children move, when. Emergency services can **plan infrastructure & cut response times** on accident-prone corridors.

A consumer app on the surface; a **public mobility dataset** underneath.

From rides to regulation & planning.

Fleet composition

Which vans & wagons dominate, and **where** — input for transport & emissions planning.

Safety hotspots

Where **unverified** & **over-capacity** vans operate — targeted child-safety regulation, not blanket crackdowns.

Fuel & emissions

Fuel consumed vs distance travelled — an economy & emissions signal optimization actively reduces.

Verification coverage, demand by neighbourhood, peak-hour pressure — all measurable for the first time.

VISION

From one family's worry to a city's safety infrastructure.

VanSafe makes the school run **safer for one child today** — and **smarter for every child** as the data grows.

Thank you.